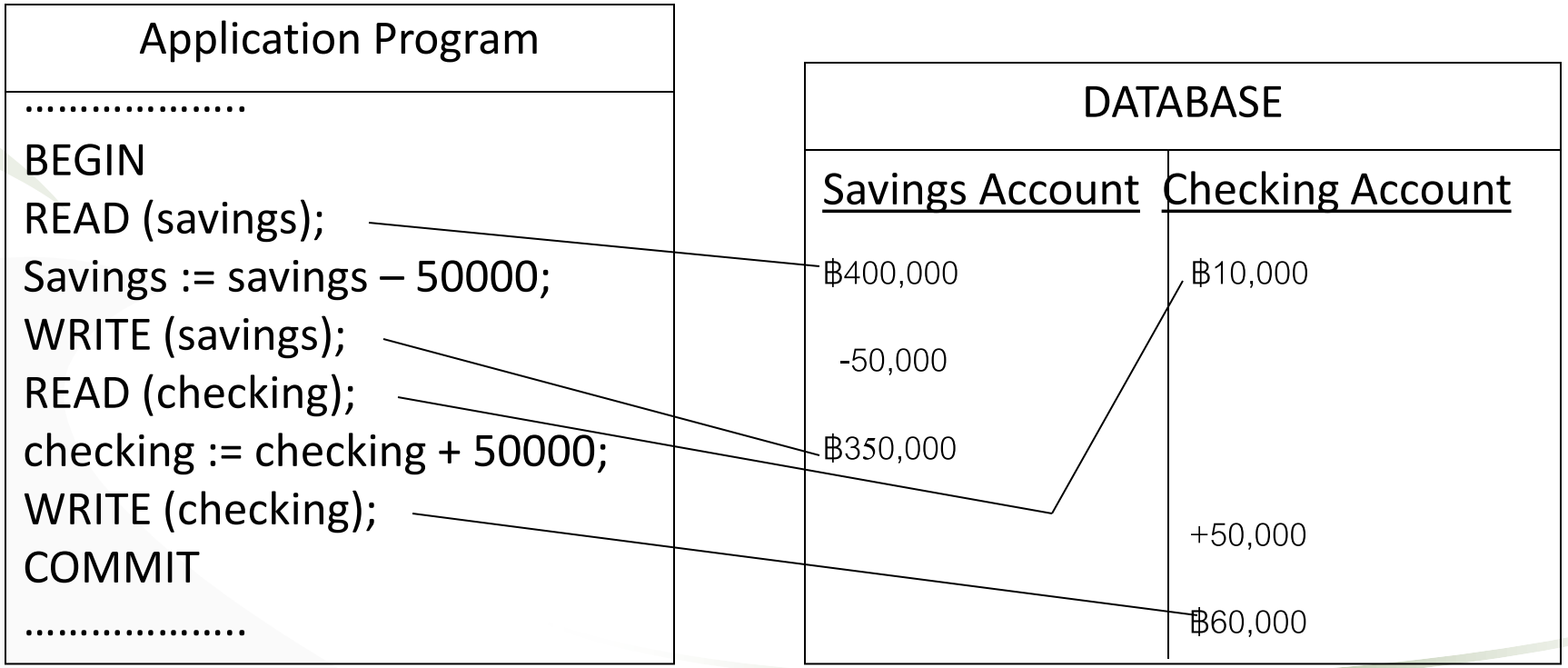
Transaction, Recovery and Concurrency Control

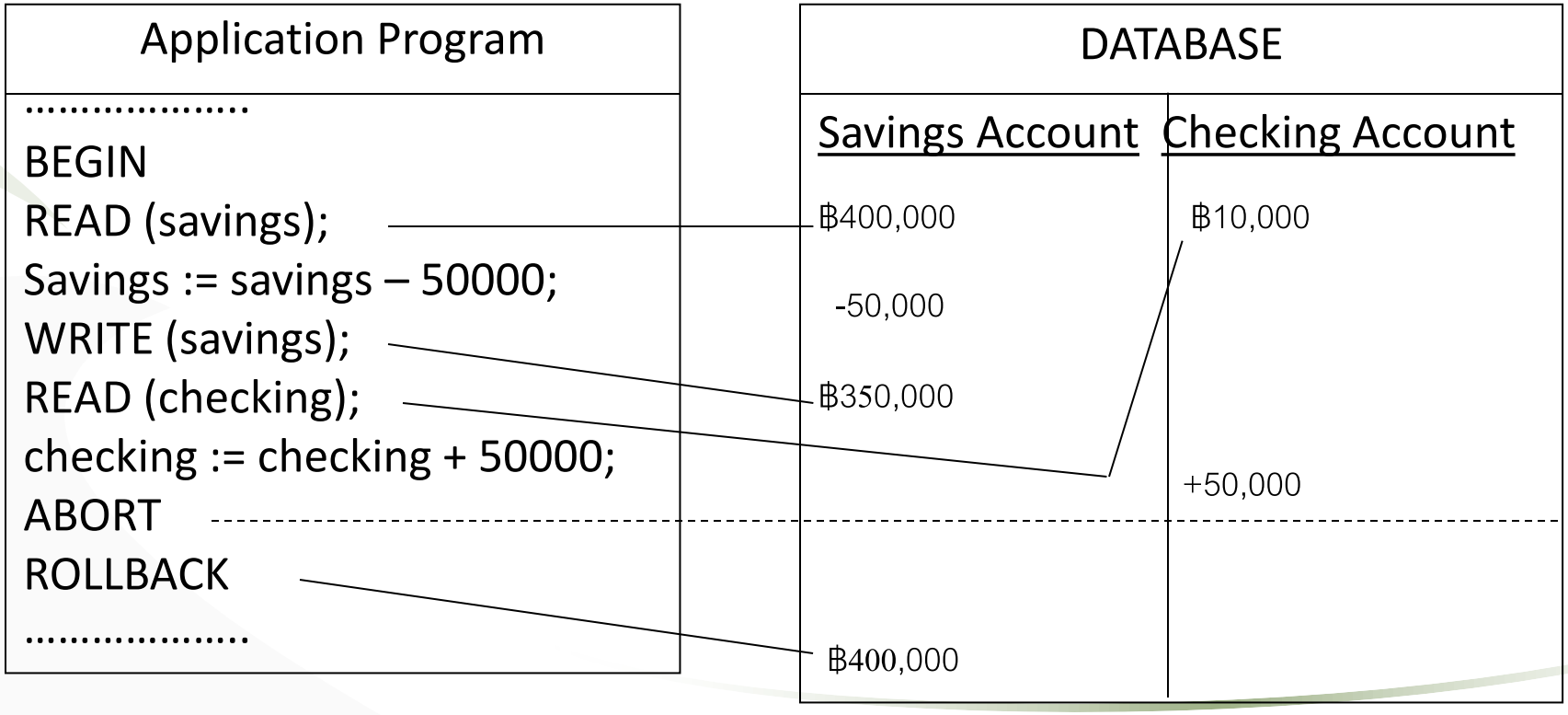
Transaction Concept

- A transaction is a unit of program execution that access and possibly updates various data items
- Example of a successful transaction



Transaction Concept (cont.)

- Example of an aborted transaction



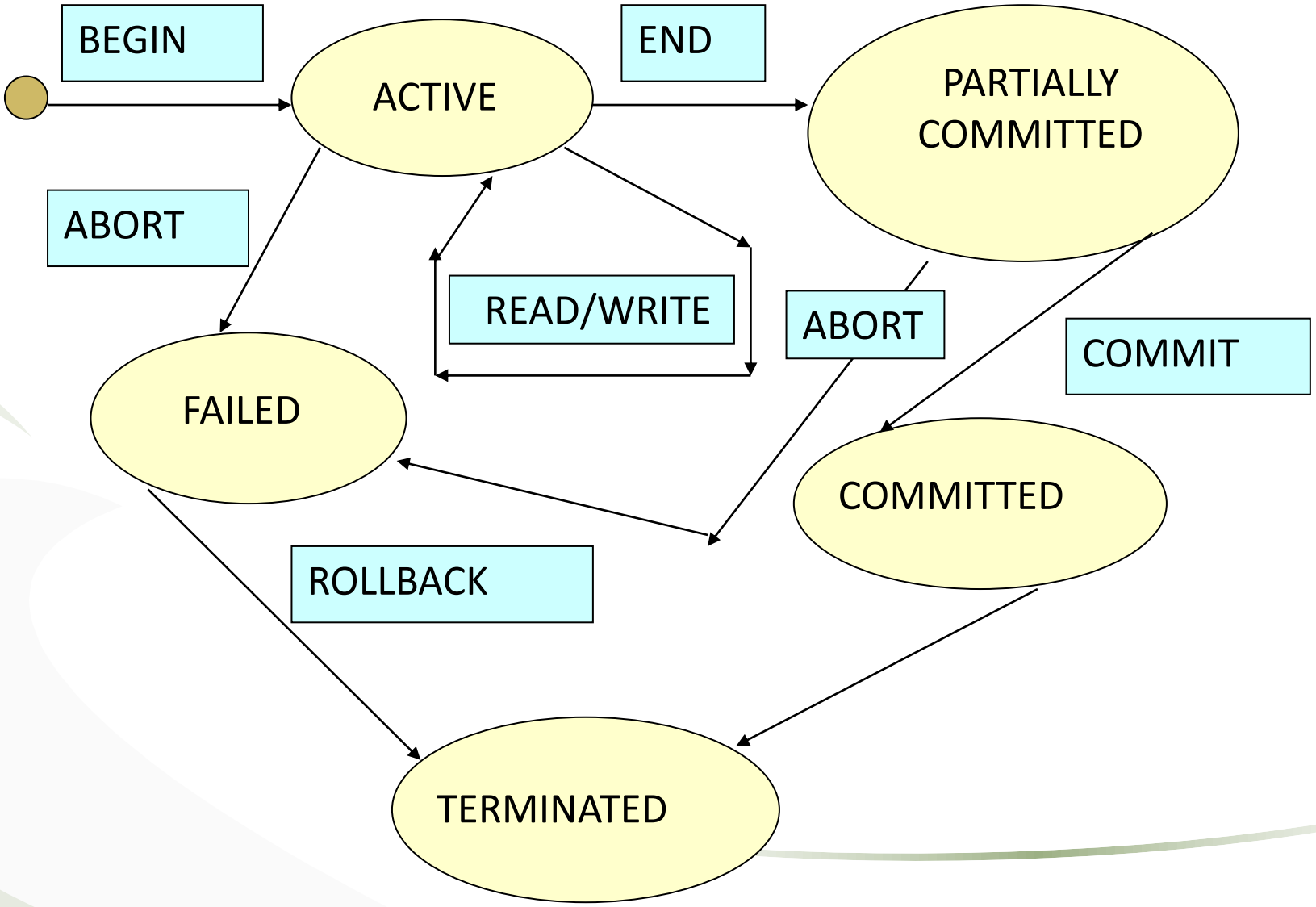
Transaction Properties

- Properties of transactions : **ACID** properties
- **Atomicity** : if it execute all of its database updates or none at all (all or nothing)
- **Consistency** : preserve database consistency
- **Isolation** : multiple transaction execute concurrency, each transaction is unaware of other transaction
- **Durability** : database update by successfully completed transactions must be permanent in the database

Transaction States

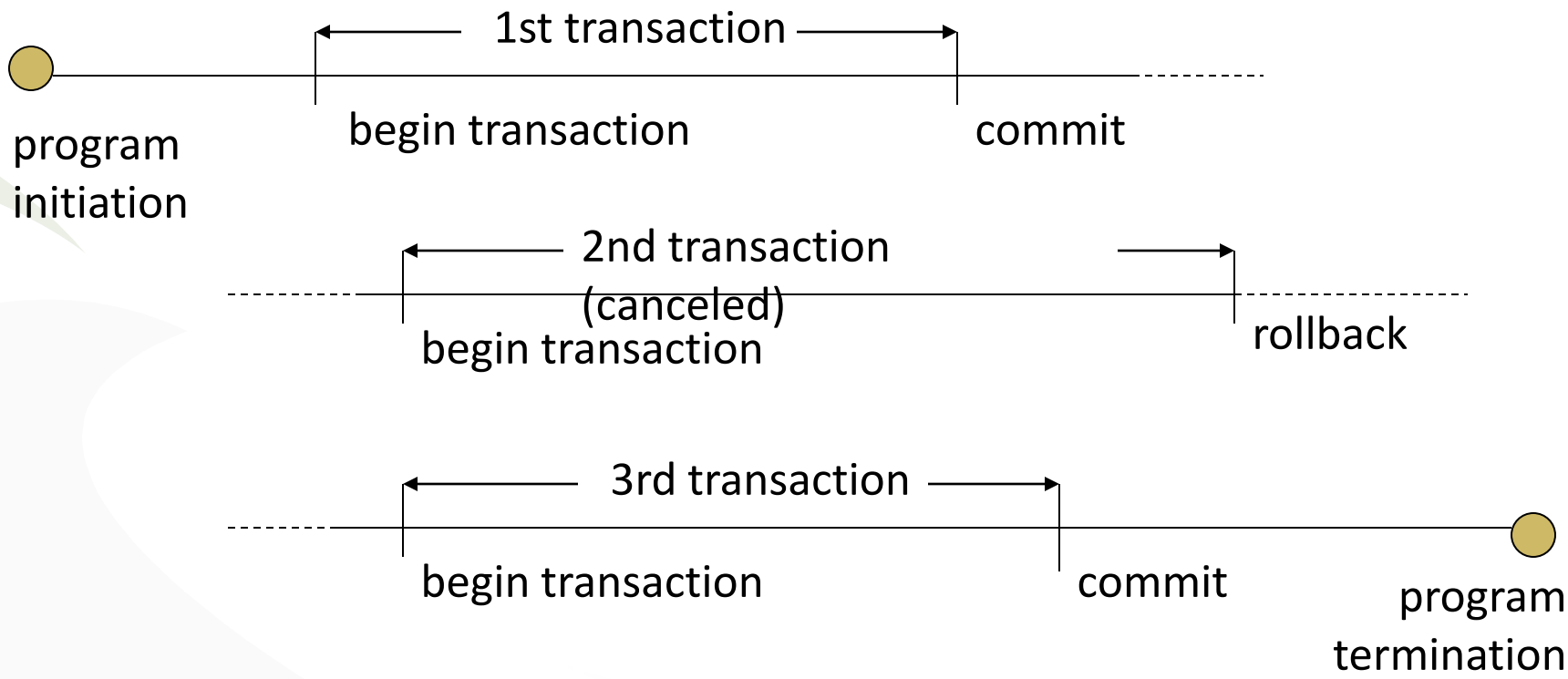
- BEGIN : transaction enters **active state**
- READ : transaction remains in **active state**
- WRITE : transaction remains in **active state**
- END : transaction proceeds to partially **committed state**
- COMMIT : transaction moves to **committed state** and finally reaches the **terminated state**
- ABORT : transaction proceeds to failed state from either **active state** or **partially committed state**
- ROLLBACK : transaction reaches the **terminated state**

Transaction States (cont.)



Recovery

- Program execution is a sequence of transactions



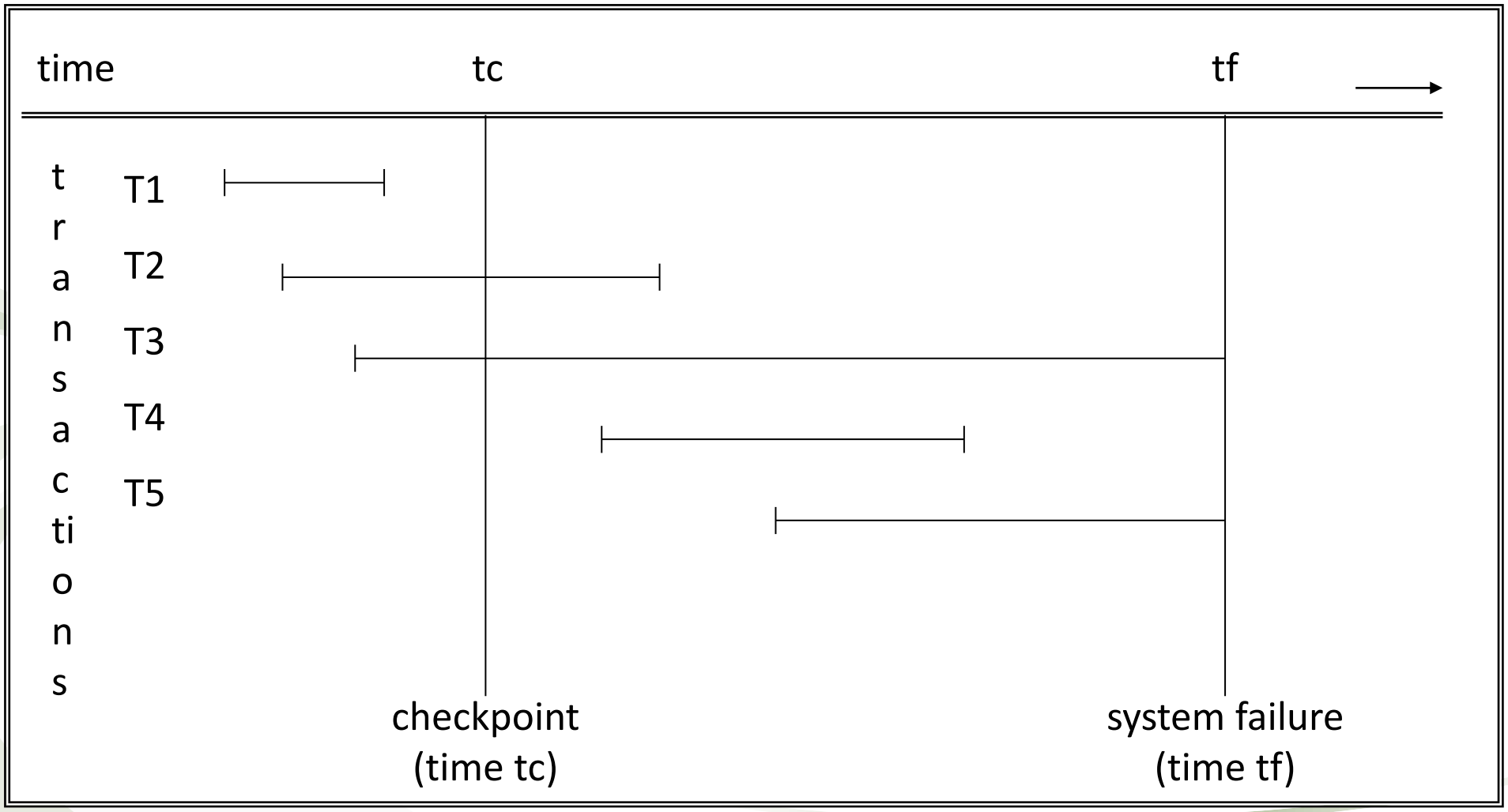
Recovery (cont.)

- Failures :

  - system failures e.g. power outage (soft crash) → system recovery (log-based recovery)

  - media failures e.g. head cash on the disk (hard crash) → media recovery

System Recovery



System Recovery (cont.)

- At restart time create UNDO/REDO list

1. Set the UNDO list equal to the list of all transactions given in the most recent checkpoint record; set the REDO list to empty.
2. Search forward through the log, start from the checkpoint record.
3. If a BEGIN TRANSACTION log entry is found for transaction T, add T to the UNDO list.
4. If a COMMIT log entry is found for transaction T, move T from the UNDO list to the REDO list.
5. When the end of the log is reached, the UNDO and REDO lists identify.

Two-phase commit

- Two-phase commit is a protocol use in the system that permit transactions to interact with two or more distinct resource manager e.g. two different DBMSs
- To maintain the transaction atomicity property

Two-phase Commit (cont.)

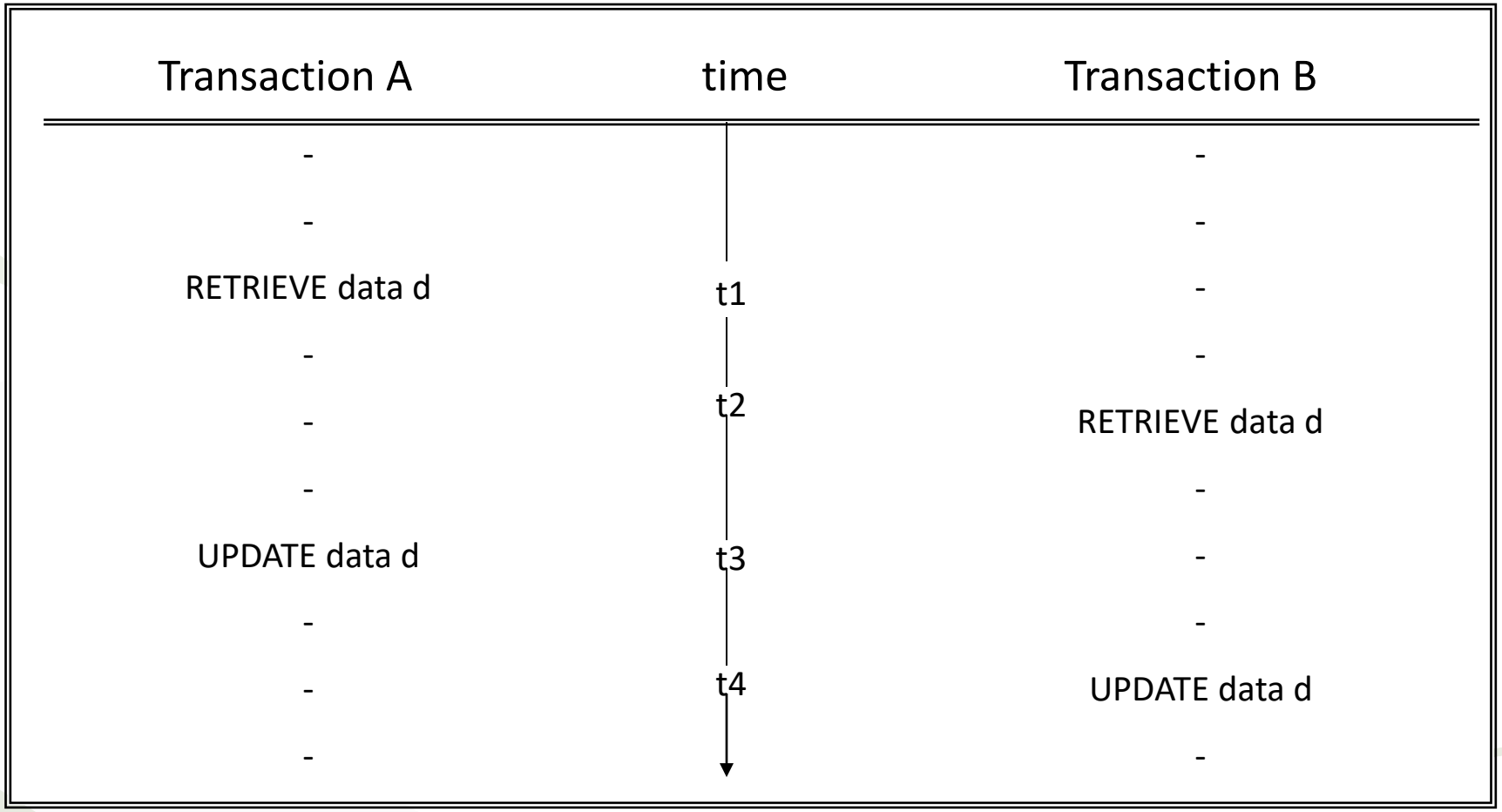
- The two phases are :-

  1. the prepare phase : the coordinator instructs all participants to “get ready to go either way”

  2. the commit phase : in which assuming all participants responded satisfactorily during the prepare phase, the coordinator the instruct all participants to perform the actual commit (or the actual rollback otherwise)

Concurrency Control

- Three concurrency problems
- The lost update problem



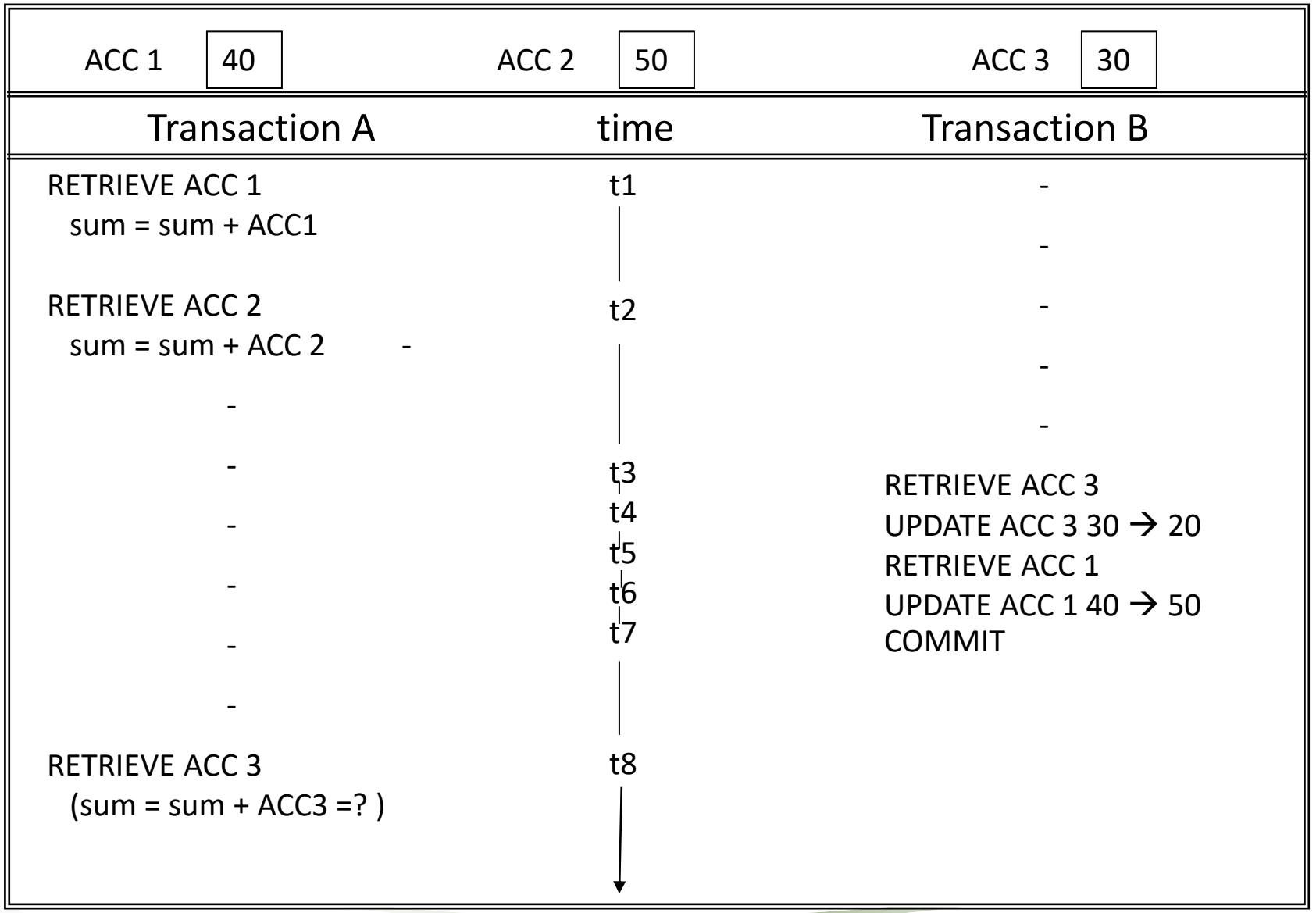
Concurrency Control (cont.)

- The uncommitted dependency problem

Transaction A	time	Transaction B
-		-
-		-
-	t1	UPDATE data d
-		-
RETRIEVE data d (or UPDATE data d)	t2	-
-		-
-	t3	ROLLBACK

Concurrency Control (cont.)

- The inconsistent analysis problem



Lock-based Resolution

- Two basic types of locks :
  1. Exclusive lock or write lock : indicated as x-lock
  2. Shared lock or read lock : indicated as s-lock
- Matrix for locking

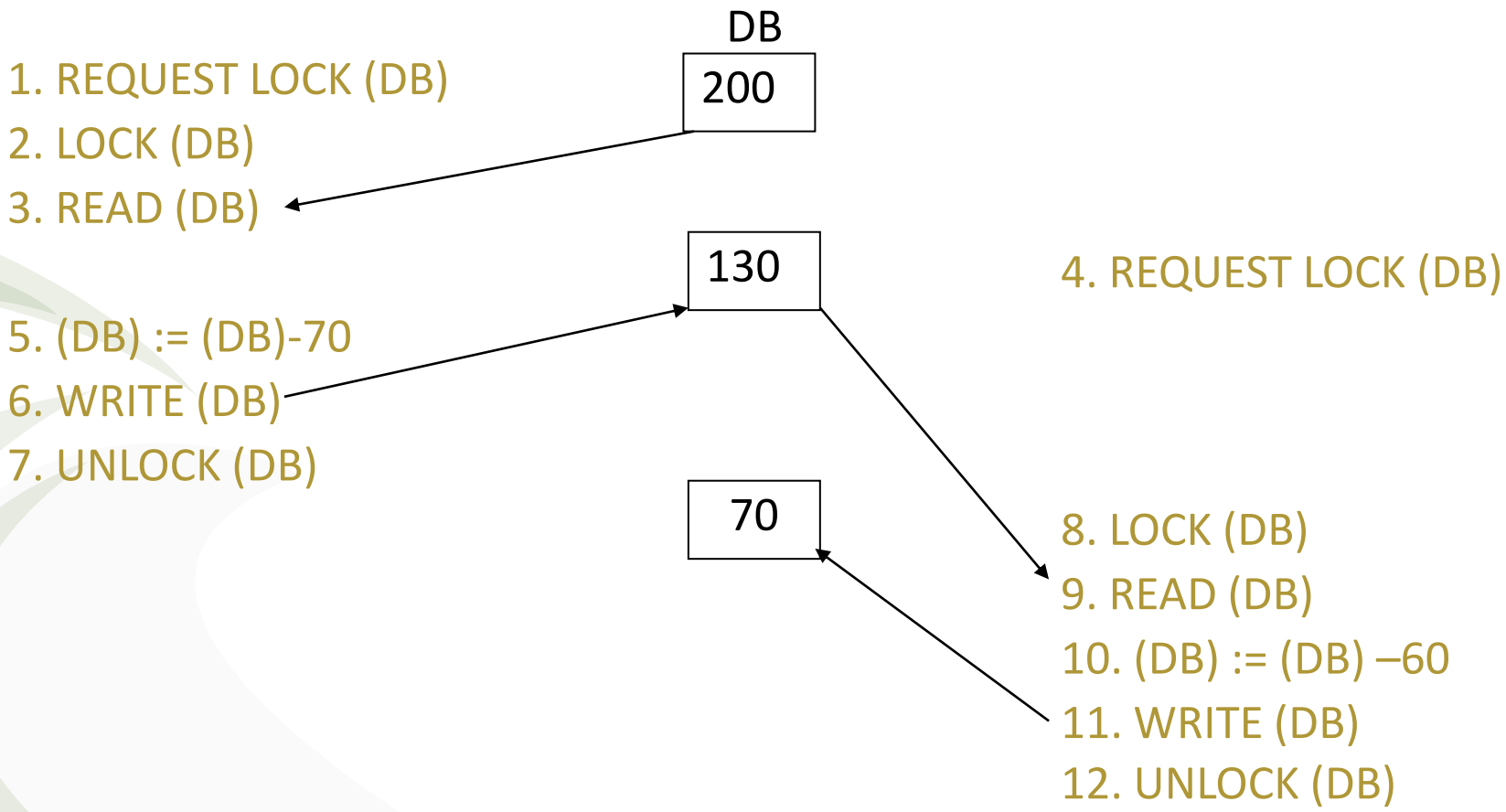
	X	S	-
X	N	N	Y
S	N	Y	Y
-	Y	Y	Y

Simple Locking

INVENTORY RECORDS IN DATATBASE

Order Entry in Chicago
(order database textbook 70 copies)

Order Entry in Boston
(order database textbook 60 copies)



Simple Locking (cont.)

- Initial value (D1) 100, (D2) 150

Operation in T1 and T2

T1	T2
LOCK-S (D2)	LOCK-S (D1)
READ (D2)	READ (D1)
UNLOCK (D2)	UNLOCK (D1)
LOCK-X (D1)	LOCK-X (D2)
READ (D1)	READ (D2)
(D1) := (D1) + (D2)	(D2) := (D2) + (D1)
WRITE (D1)	WRITE (D2)
UNLOCK (D1)	UNLOCK (D2)
…	…

Simple Locking (cont.)

TIME	T1 Chicago order	T2 Boston order	DB
	DB 70 Copies	DB 60 Copies	
1	BEGIN T1		200
2	LOCK-X (DB)	BEGIN T2	200
3	READ (DB)	LOCK-X (DB)	200
4	(DB) := (DB) - 70	…. wait ….	200
5	WRITE (DB)	…. wait ….	130
6	UNLOCK (DB)	…. wait ….	130
7	COMMIT	READ (DB)	130
8	END	(DB) := (DB) – 60	130
9		WRITE (DB)	70
10		UNLOCK (DB)	70
11		COMMIT	70
12		END	70